PARENT ONLY — SOLVED ANSWER KEY

Vyom · Parent answer key

30 Nov – 6 Dec 2026 · Monday–Sunday · NDPS Class 9

Vyom ko test se pehle ye PDF mat dena. Student pack mein sirf questions aur blank answer space hai. Is alag parent PDF mein har original question ke saath uska matching solved answer, working aur marking points diye hain; Test ID se student paper match karein.

Checking: same evening. C = concept, R = recall, K = careless, T = time.

90%+: short revision. 75–89%: weak-topic practice + retest. Below 75%: topic dobara samjhao; mistakes notebook mein likho.

Answer keys existing source question banks se li gayi hain; school pattern abhi unconfirmed hai. Alternate valid answers ko marking guidance ke mutabik accept karein.

Week score register

Date / test ID	Test	Marks	Galti / retest
30 Nov T2026-11-30	AI Weekly test	___ / 40	Retest: 3 Dec
1 Dec D2026-12-01	Daily mixed Daily test	___ / 10	Retest: 4 Dec
2 Dec D2026-12-02	Daily mixed Daily test	___ / 10	Retest: 5 Dec
3 Dec D2026-12-03	Daily mixed Daily test	___ / 10	Retest: 6 Dec
4 Dec D2026-12-04	Daily mixed Daily test	___ / 10	Retest: 7 Dec
5 Dec D2026-12-05	Daily mixed Daily test	___ / 10	Retest: 8 Dec
6 Dec D2026-12-06	Daily mixed Daily test	___ / 10	Retest: 9 Dec

Artificial Intelligence (417) · Weekly Test 1 · Answer key

Monday, 30 November 2026 · 40 marks

Test ID: T2026-11-30

Class 9 · Part A: Communication, Self-Management, ICT, Entrepreneurial Skills · Part B: AI Reflection, Project Cycle & Ethics, Data Literacy, Math for AI

Q1 **[1]**

Question: Which of these is NOT one of the 7 Cs of effective communication?

(a) Clear
(b) Concise
(c) Complicated
(d) Courteous

Expected answer / marking points: (c)

Q2 **[1]**

Question: Facial expressions, gestures and eye contact are examples of:

(a) verbal communication
(b) written communication
(c) non-verbal communication
(d) visual aids only

Expected answer / marking points: (c)

Q3 **[1]**

Question: The correct order of the communication cycle is:

(a) Sender → Receiver → Message → Feedback
(b) Sender → Message → Channel → Receiver → Feedback
(c) Message → Sender → Feedback → Receiver
(d) Receiver → Channel → Sender → Message

Expected answer / marking points: (b)

Q4 **[1]**

Question: Which is a healthy way to manage exam stress?

(a) Studying all night without sleep
(b) Regular exercise, enough sleep and a study timetable
(c) Skipping meals to save time
(d) Avoiding all breaks

Expected answer / marking points: (b)

Q5 **[1]**

Question: Knowing your own strengths and weaknesses is called:

(a) self-awareness
(b) self-confidence
(c) teamwork
(d) time management

Expected answer / marking points: (a)

Q6 **[1]**

Question: The domain of AI that works with images and videos is:

(a) Natural Language Processing
(b) Computer Vision
(c) Statistical Data
(d) Robotics only

Expected answer / marking points: (b)

Q7 **Question:** A voice assistant that understands spoken questions mainly uses:
[1]

(a) Computer Vision
(b) Natural Language Processing
(c) Spreadsheets
(d) Data Exploration

Expected answer / marking points: (b)

Q8 **Question:** The first stage of the AI Project Cycle is:
[1]

(a) Data Acquisition
(b) Modelling
(c) Problem Scoping
(d) Evaluation

Expected answer / marking points: (c)

Q9 **Question:** The 4Ws Problem Canvas consists of:
[1]

(a) Who, What, Where, Why
(b) Who, When, Which, Why
(c) What, When, Where, How
(d) Why, How, Who, Whom

Expected answer / marking points: (a)

Q10 **Question:** A System Map is used to show:
[1]

(a) the hardware of a computer
(b) relationships (cause and effect) between elements of a system
(c) only the final output of a model
(d) the marks of students

Expected answer / marking points: (b)

Q11 **Question:** Which of these is a source of data collection?
[1]

(a) Surveys
(b) Monitor
(c) Keyboard shortcut
(d) Printer

Expected answer / marking points: (a)

Q12 **Question:** An AI system trained mostly on data of one group gives unfair results for others. This problem is called:
[1]

(a) AI bias
(b) AI access
(c) Data exploration
(d) Deployment

Expected answer / marking points: (a)

Q13 **Question:** Data arranged in rows and columns, like a spreadsheet, is:
[1]

(a) unstructured data
(b) structured data
(c) image data
(d) audio data

Expected answer / marking points: (b)

Q14 [1]

Question: The mean of 4, 6, 8, 10, 12 is:

(a) 6 (b) 8

(c) 10 (d) 40

Expected answer / marking points: (b) 40 ÷ 5 = 8

Q15 [1]

Question: A fair die is rolled once. The probability of getting an even number is:

(a) 1/6 (b) 1/3

(c) 1/2 (d) 2/3

Expected answer / marking points: (c) 3/6

Q16 [1]

Question: Which part of a computer is often called its "brain"?

(a) Monitor (b) RAM

(c) CPU (d) Hard disk

Expected answer / marking points: (c)

Q17 [2]

Question: Differentiate between verbal and non-verbal communication with one example of each.

Expected answer / marking points: Verbal uses words — spoken or written (phone call, letter) [1]; non-verbal uses gestures, expressions, posture, eye contact (smile, thumbs-up) [1].

Q18 [2]

Question: Name two barriers to effective communication and suggest one way to overcome each.

Expected answer / marking points: Any two: language barrier — use simple words; noise/physical — move to a quiet place; emotional (anger/fear) — stay calm, listen; cultural differences — be respectful [1 + 1].

Q19 [2]

Question: In the 4Ws Problem Canvas, what do the "Who" and "Why" blocks help us find out?

Expected answer / marking points: Who: the stakeholders — people affected by the problem and what we know about them [1]. Why: the benefits of solving the problem for the stakeholders and how it fits their needs [1].

Q20 [2]

Question: Give two examples of AI in daily life and name the AI domain each belongs to.

Expected answer / marking points: e.g., Face unlock on phone — Computer Vision; Voice assistant/chatbot — NLP; Movie or shopping recommendations — Statistical Data (any two) [1 + 1].

Q21 [2]

Question: Differentiate between qualitative and quantitative data with one example each.

Expected answer / marking points: Qualitative: describes qualities/categories, not measured in numbers (favourite colour, feedback) [1]. Quantitative: numerical, can be counted/measured (height, marks) [1].

Q22 [2]

Question: Find the median and mode of: 5, 3, 8, 3, 9, 6, 3, 10.

Expected answer / marking points: Sorted: 3, 3, 3, 5, 6, 8, 9, 10. Median = (5 + 6) ÷ 2 = 5.5 [1]; Mode = 3 (appears 3 times) [1].

Q23 [4] **Question:** Explain the stages of the AI Project Cycle using the example of "predicting which students may need extra help before exams".

Expected answer / marking points: Problem Scoping — define problem with 4Ws (who needs help, why) [0.75]; Data Acquisition — collect test marks, attendance, homework records [0.75]; Data Exploration — make graphs to find patterns [0.75]; Modelling — choose a model/rule that predicts risk [0.75]; Evaluation — test accuracy on new data; (Deployment — use it in school) [1].

Q24 [4] **Question:** Explain any two ethical concerns in AI (such as bias and privacy) with an example of each, and suggest how each can be reduced.

Expected answer / marking points: Bias: data from one group → unfair results (e.g., face recognition works poorly for some skin tones); reduce by diverse, balanced data and testing [2]. Privacy: personal data collected without consent (e.g., apps tracking location); reduce by consent, minimum data collection, security [2].

Q25 [4] **Question:** Describe any four qualities of a successful entrepreneur, with a one-line example for each.

ATHVA / OR

Explain four steps of good time management that a Class 9 student can follow.

Expected answer / marking points: Any four: risk-taking, creativity/innovation, hard work and persistence, decision-making, leadership, confidence, customer focus — each with example [1 × 4].
OR: Make a to-do list / timetable; set priorities (urgent vs important); break big tasks into small steps; avoid distractions like phone; review at day-end; include breaks and sleep (any four) [1 × 4].

Score: ___ / 40 Checked by: _____

Weak topic aur correction:

Daily Test · Answer key

Tuesday, 1 December 2026 · 10 marks

Test ID: D2026-12-01

Maths mixed revision · Science mixed revision · AI Math for AI · Maths Ch 13 Two Variables, One Line

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Q1 **[3]** **Question:** The graph of a linear polynomial p(x) passes through (2, 3) and (6, 11). Find p(x) and the point where its graph cuts the x-axis.

Expected answer / marking points: Slope = (11 – 3)/(6 – 2) = 2 [1]; 3 = 2(2) + b ⇒ b = –1, p(x) = 2x – 1 [1]; x-axis: (1/2, 0) [1].

Q2 **[1]** **Question:** Are the lines y = 4x + 1 and y = 4x – 3 parallel? Why?

Expected answer / marking points: Yes — both have the same slope 4 (different y-intercepts).

Q3 **[2]** **Question:** Differentiate between a scientific law and a scientific theory with one example each.

Expected answer / marking points: Law describes a regular pattern (e.g., Newton's laws) [1]; theory explains why the pattern occurs, based on evidence (e.g., atomic theory) [1].

Q4 **[2]** **Question:** A bag has 4 red and 6 green balls. Find the probability of picking a green ball.

Expected answer / marking points: 6/10 [1] = 3/5 [1].

Q5 **[2]** **Question:** A line passes through (0, 3) and has slope 2. Write its equation and find y when x = 4.

Expected answer / marking points: y = 2x + 3 [1]; y = 11 [1].

Score: ___ / 10 Checked by: _____

Weak topic aur correction:

Daily Test · Answer key

Wednesday, 2 December 2026 · 10 marks

Test ID: D2026-12-02

SST mixed revision · Hindi mixed revision

Q1 [1]

Question: Name the four disciplines of Social Science studied in Grade 9.

Expected answer / marking points: History, Geography, Political Science, Economics.

Q2 [1]

Question: Which discipline studies how people make choices about scarce resources?

Expected answer / marking points: Economics.

Q3 [2]

Question: Why do we need to study Social Science? Give two reasons.

Expected answer / marking points: Helps understand society, past and present [1]; makes us informed, responsible citizens who can solve real problems (any two) [1].

Q4 [2]

Question: How are History and Geography connected? Give an example.

Expected answer / marking points: Geography (rivers, soil, climate) shaped where and how people lived [1]; e.g., early civilisations grew in fertile river valleys [1].

Q5 [1]

Question: 'दो बैलों की कथा' के लेखक कौन हैं?

Expected answer / marking points: प्रेमचंद।

Q6 [1]

Question: हीरा और मोती किसके बैल थे?

Expected answer / marking points: झूरी के।

Q7 [2]

Question: गया के घर हीरा-मोती के साथ कैसा व्यवहार हुआ?

Expected answer / marking points: उन्हें सूखा भूसा दिया गया, भारी काम लिया गया और मारा गया [1]; वे इसे अपना अपमान समझकर वहाँ से भाग निकले [1]।

Score: ___ / 10 Checked by: _____

Weak topic aur correction:

Daily Test · Answer key

Thursday, 3 December 2026 · 10 marks

Test ID: D2026-12-03

English mixed revision · AI mixed revision

Q1 **Question:** Why did Avva touch the narrator's feet? What does it show about her?
[2]

Expected answer / marking points: She was honouring her teacher, not her granddaughter [1]; shows her humility and respect for learning/teachers [1].

Q2 **Question:** Which craft did Arenla want Sentila to learn?
[1]

Expected answer / marking points: Weaving.

Q3 **Question:** Why did Arenla not want Sentila to become a pot maker?
[2]

Expected answer / marking points: Pot making was hard — clay had to be carried 16 km uphill and pounded, and pots took months [1]; it earned only a pittance, while weaving paid more and could be done indoors [1].

Q4 **Question:** Give two ways to build self-confidence.
[2]

Expected answer / marking points: Set small goals and achieve them; positive self-talk; practise skills; learn from mistakes (any two) [2].

Q5 **Question:** Make a SMART goal for improving your Maths marks.
[3]

Expected answer / marking points: Example: "I will solve 10 Maths questions daily for 6 weeks to raise my score from 70% to 85% by the February mock" — specific, measurable, achievable, relevant, time-bound [3].

Score: ___ / 10 Checked by: _____

Weak topic aur correction:

Daily Test · Answer key

Friday, 4 December 2026 · 10 marks

Test ID: D2026-12-04

Maths mixed revision · Science mixed revision

Q1 **[2]** **Question:** In a matchstick pattern of joined hexagons, stage n needs 5n + 1 matchsticks. How many are needed for stage 10? Can 100 matchsticks form a stage exactly?

Expected answer / marking points: Stage 10: 51 [1]. 5n + 1 = 100 ⇒ n = 19.8, not whole, so no [1].

Q2 **[2]** **Question:** The diameter of a circle is 26 cm and a chord of length 24 cm is drawn. Find the distance of the chord from the centre.

Expected answer / marking points: r = 13, half-chord = 12; distance = √(169 − 144) = 5 cm [1 + 1].

Q3 **[2]** **Question:** In a circle of radius 13 cm, a chord is 5 cm away from the centre. Find the length of the chord.

Expected answer / marking points: Half-chord = √(169 − 25) = 12 cm; chord = 24 cm [1 + 1].

Q4 **[1]** **Question:** Ligaments connect _____ to _____.

Expected answer / marking points: bone to bone

Q5 **[2]** **Question:** Why are the epithelial tissues lining the lungs only one or a few cells thick?

Expected answer / marking points: A thin layer allows quick exchange (diffusion) of gases between air and blood [2].

Q6 **[1]** **Question:** What is a hypothesis?

Expected answer / marking points: A testable, possible explanation for an observation.

Score: ___ / 10 Checked by: _____

Weak topic aur correction:

Daily Test · Answer key

Saturday, 5 December 2026 · 10 marks

Test ID: D2026-12-05

SST mixed revision · Hindi mixed revision

Q1 [1]

Question: Name the agent of gradation that forms a meander and a delta.

Expected answer / marking points: Running water (river).

Q2 [2]

Question: Differentiate between weathering and erosion.

Expected answer / marking points: Weathering: breaking of rocks in place [1]; erosion: wearing away and carrying away of material by agents like water, wind, ice [1].

Q3 [1]

Question: What are microliths?

Expected answer / marking points: Very small stone tools of the Mesolithic period.

Q4 [2]

Question: Give two features of the Neolithic period.

Expected answer / marking points: Farming and domestication of animals; permanent villages; pottery and weaving; polished stone tools (any two) [2].

Q5 [2]

Question: इस कहानी से हमें कौन-सी दो सीखें मिलती हैं?

Expected answer / marking points: स्वतंत्रता के लिए संघर्ष करना चाहिए [1]; सच्ची मित्रता, एकता और अन्याय का विरोध (कोई एक और) [1]।

Q6 [1]

Question: 'क्या लिखूँ' निबंध के लेखक कौन हैं?

Expected answer / marking points: पदुमलाल पुन्नालाल बख्शी।

Q7 [1]

Question: नमिता और अमिता ने लेखक से किन विषयों पर निबंध लिखने को कहा?

Expected answer / marking points: 'दूर के ढोल सुहावने होते हैं' और 'समाज-सुधार'।

Score: ___ / 10 Checked by: _____

Weak topic aur correction:

Daily Test · Answer key

Sunday, 6 December 2026 · 10 marks

Test ID: D2026-12-06

English mixed revision · Maths mixed revision

Q1 **[2]** **Question:** What did the village council say about skills like pot making?

Expected answer / marking points: Such skills do not belong to any individual; they carry the tradition of the people [1]; experts must pass them on to anyone who wishes to learn [1].

Q2 **[2]** **Question:** How can the craft of pankha-making be preserved?

Expected answer / marking points: Celebrating pankhas and their culture; workshops and handicraft exhibitions [1]; giving artisans a commercial platform for a sustainable livelihood [1].

Q3 **[1]** **Question:** What is “Vitamin-M” in the story?

Expected answer / marking points: An imaginary vitamin for memory that Ravi’s mother wished someone would invent.

Q4 **[1]** **Question:** Give a counterexample to: “Every number divisible by 2 is divisible by 4.”

Expected answer / marking points: 6 (or 2, 10, 14 ...) — divisible by 2 but not by 4.

Q5 **[2]** **Question:** A T20 team’s run rate after 19 overs is 6. It scores 12 runs in the 20th over. What is the run rate after 20 overs?

Expected answer / marking points: Runs after 19 overs = 114 [1]; total 126 ÷ 20 = 6.3 [1].

Q6 **[2]** **Question:** Class A has 30 students with an average of 60 marks; Class B has 20 students with an average of 70. Find the average of all 50 students.

Expected answer / marking points: (1800 + 1400) ÷ 50 = 64 [method 1 + answer 1]. (Not 65 — simple average of averages is wrong for unequal groups.)

Score: ___ / 10 Checked by: _____

Weak topic aur correction:

Parent Sunday handover checklist

- [ ] Agle Monday–Sunday ka student pack print karke diya.
- [ ] Parent solved key alag rakhi; student copy ke saath share nahi ki.
- [ ] Pichhle week ka pending kaam aur 3 weak topics review kiye.
- [ ] School exam date sheet aur tuition timings se schedule match kiya.

Agli priority / teacher se poochhna: